

AMENDMENT NO. 1 JUNE 2009
TO
IS 15642 (PARTS 1 AND 2) : 2006 SPECIFICATION FOR QUICK METHODS
FOR DETECTION OF ADULTERANTS/CONTAMINANTS IN
COMMON FOOD PRODUCTS

(Second cover, Foreword, para 3, line 3) — Substitute ‘consumer organizations’ for ‘household’.

(Second cover, Foreword, para 4, line 2) — Substitute ‘consumers’ for ‘housewives’.

(Second cover, Foreword, para 4, line 4) — Substitute ‘consumers’ for ‘housewives’.

(Page 1, clause 2, Title) — Substitute ‘**APPARATUS/CHEMICALS**’ for ‘**APPARATUS**’.

[Page 1, clause 2(j)] — Delete the word ‘Milk’.

(Page 1, clause 2) — Insert the following after (j):

‘k) Common salt (*see* IS 253)’.

(Page 1, clause 3.2) — Substitute ‘**Detection of Ergot Affected Grains**’ for ‘**Detection of Ergot Effected Grains**’.

(Page 1, clause 3.1.1) — Substitute the following for the existing:

‘Foodgrains meant for human consumption shall be whole or broken kernels of cereals and pulses. Small stones, marble chips identical to foodgrains and pulses are mixed or get mixed, if good agricultural practices are not followed. These inorganic foreign matters may irritate the mucus membranes of human digestive system. Impurities of animal origin are also found in foodgrains, if not properly stored. Similarly, the excreta of animal origin may carry harmful bacteria.’

(Page 1, clause 3.3.1, line 7) — Substitute ‘yellow or light’ for ‘yellow and light’.

[Page 5, clause 2(f)] — Delete the word ‘Milk’.

[Page 5, clause 2(g)] — Delete the word ‘and’.

(Page 5, clause 2) — Insert the following after (h):

‘j) Dropper;
k) Water bath;
m) Spatula; and
n) Spot Tile.’

[Page 5, clause 3(b), line 3] — Substitute ‘ethyl alcohol’ for ‘alcohol’.

[Page 5, clause 3(m), line 2] — Substitute ‘12.7’ for ‘12.69’ and substitute ‘potassium iodide’ for ‘potassium iodine’.

[Page 5, clause 3(p)] — Substitute ‘Petroleum ether (60-80°C)’ for ‘Petroleum ether’.

[Page 5, clause 3(w)] — Insert the following after (w):

‘y) Resorcinol powder;
z) Sodium Hydroxide — 2 percent;
aa) Sugar crystals — about 25 g;
bb) Sulphuric Acid — 10 percent; and
cc) Zinc chloride.’

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(Page 5, clause 4.1.1, line 4) — Insert ‘(rosin)’ for ‘Colophony’.

(Page 8, clause 4.18.2) — Insert the following clauses after 4.18.2:

4.19 Detection of Date and Tamarind Seed Powder in Coffee

4.19.1 Procedure

Sprinkle a small amount of the test sample on a filter or blotting paper. Add dropwise 2 percent sodium hydroxide solution so as to completely wet the coffee powder. Coffee particles give yellow colour. The appearance of red colour indicates the presence of date seed powder while the appearance of pink spots or trails indicates the presence of tamarind seed powder.

4.20 Detection of Cane Sugar in Milk/Ice-Cream

4.20.1 General

Cane sugar may be added to milk to raise the density to prevent detection of extraneous water.

4.20.2 Procedure

There are two methods for the detection of cane sugar in milk.

4.20.2.1 Method I

Take 10 ml of milk sample in a test tube. Add 1 ml of concentrated hydrochloric acid and mix well. Add 100 mg of resorcinol powder and mix thoroughly. Place the tube in boiling water-bath and observe the colour. The appearance of a red colour indicates the presence of cane sugar.

4.20.2.2 Method II

Dissolve 1 g of resorcinol in 100 ml of (1:1.5) hydrochloric acid. Curdle an aliquot of milk by adding a little concentrated hydrochloric acid (1 ml for 25 ml sample). Allow to stand for 10 min and filter. Take 5 ml of the reagent in a test tube, and add 1 ml of filtered milk serum and mix well. Place the tube in boiling water for 1 min and observe the colour. The appearance of a red colour indicates the presence of cane sugar.

4.21 Detection of Cellulose in Milk/Ice-Cream

4.21.1 General

This adulterant is soluble in a concentrated solution of zinc chloride and gives a blue colour with iodine zinc chloride reagent.

4.21.2 Reagent

4.21.2.1 Reagent A

Dissolve 20 g of zinc chloride in 8.5 ml of water and cool.

4.21.2.2 Reagent B

Dissolve 3 g of potassium iodide and 1.5 g of iodine in 60 ml of water.

4.21.3 Procedure

Add Reagent B to Reagent A drop by drop until iodine begins to precipitate. Take 10 ml of milk in a 100-ml beaker. Add 50 ml of hot water and stir thoroughly for about 2 min. Filter the mixture through nylon

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cloth. Wash the residue twice with 50 ml portions of hot water. Transfer the residue with a spatula to a spot tile. Stain one part of residues with Reagent A and another part with Reagent B. Development of blue colour with Reagent A and absence of blue colour with Reagent B solution confirms presence of cellulose.

(Page 9, Annex A) — Insert the following after (xxiii):

<i>Sl No.</i>	<i>Common Name</i>	<i>Botanical Name</i>
(1)	(2)	(3)
'xxiv)	Castor	<i>Ricinus communis</i>